Journal of SME Research

Perception of AI Implementation in Creative Fields among Employees of Small and Medium Enterprises

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This manuscript was compile on October 25, 2024

Abstract

This study investigates the perceptions of employees in small and medium enterprises (SMEs) regarding the implementation of artificial intelligence (AI) in creative fields. The research aims to explore how AI tools influence job performance, creativity, and overall job satisfaction. A mixed-methods approach, combining qualitative interviews and quantitative surveys, was utilized to gather comprehensive insights. The findings suggest a nuanced relationship between AI implementation and employee perceptions, highlighting both potential benefits and concerns related to job security and creativity.

Keywords: *AI implementation, SMEs, Creative fields, Employee perceptions, Job satisfaction*

1. Introduction

The implementation of artificial intelligence (AI) has increasingly permeated various sectors, particularly in creative fields such as design, marketing, and content creation. This research aims to examine how employees of small and medium enterprises (SMEs) perceive the adoption of AI technologies in their work environments. Understanding these perceptions is crucial as it can influence both job satisfaction and performance outcomes. Previous studies indicate mixed reactions from employees regarding AI, where potential efficiency gains are often tempered by concerns about job security and the erosion of creative skills.

2. Literature Review

The integration of Artificial Intelligence (AI) into the workplace has become increasingly prevalent, particularly in Small and Medium-sized Enterprises (SMEs) operating in creative sectors. Understanding employee perceptions of AI implementation is critical, as it influences job satisfaction, performance, and overall workplace dynamics. This literature review explores existing research on how employees in SMEs perceive AI integration in creative fields, examining factors such as job satisfaction, trust in technology, ethical concerns, and the impact on creativity.

2.1. The Role of AI in Creative Fields

AI technologies have emerged as valuable tools in creative industries, from content creation and graphic design to marketing and customer engagement. These tools can enhance productivity, streamline workflows, and provide data-driven insights. For instance, a study in 2021 highlights that AI can augment human creativity by generating ideas, allowing employees to focus on higher-level tasks. [2] However, while AI can enhance efficiency, it also raises concerns regarding its implications for job roles traditionally held by humans, particularly in creative sectors.

2.2. Employee Perceptions and Job Satisfaction

The perception of AI among employees in SMEs significantly impacts their job satisfaction. A Research in 2023 indicates that employees who view AI as a complementary tool rather than a replacement for their skills report higher job satisfaction levels. Conversely, fear of job displacement can lead to anxiety and dissatisfaction [5]. This dichotomy suggests that effective communication about the role of AI in enhancing rather than diminishing creative capabilities is essential.

Moreover, the study by Ploin A. in 2022 emphasizes the importance of involving employees in the AI implementation process. When employees are engaged in decision-making and feel their opinions are valued, they are more likely to perceive AI positively. This involvement can mitigate feelings of uncertainty and resistance to change, fostering a more supportive work environment [3].

2.3. Trust and Acceptance of AI Technologies

Trust in AI technologies plays a crucial role in shaping employee perceptions. Research by Amato in 2019 found that employees who have a higher level of trust in AI systems are more likely to embrace their use in creative tasks. This trust can be influenced by various factors, including previous experiences with technology, the perceived reliability of AI tools, and the clarity of their intended use [1]. For SMEs, where resources may be limited, investing in training and education about AI can enhance trust and acceptance among employees.

Conversely, lack of trust can lead to skepticism and resistance. A study by Epstein and Ziv in 2023 illustrates that employees who perceive AI as opaque or difficult to understand are less likely to accept its use in their workflows. This highlights the necessity for transparent communication regarding how AI systems function and the rationale behind their implementation [4].

2.4. Ethical Concerns and Impact on Creativity

Ethical concerns surrounding AI implementation are particularly salient in creative fields. Employees often express worries about the ethical implications of using AI in content creation, such as issues related to originality, authorship, and the potential for biased outputs. Studies show that many employees in creative SMEs are apprehensive about the consequences of relying on AI-generated content, fearing it may undermine human creativity and lead to homogenization of artistic expression [5].

Additionally, research indicates that while AI can assist in generating ideas, over-reliance on technology may stifle individual creativity. A study in 2021 found that employees who rely heavily on AI tools reported decreased motivation and creativity in their work. This suggests a potential paradox where, although AI is intended to support creative processes, it may inadvertently hinder the very essence of creativity by diminishing personal investment in the work [2].

3. Methodology

This study employs a mixed-methods approach, combining both qualitative and quantitative research techniques, however in this particular research, qualitative research was the most focused on as the topic revolves around the perception of employees on the implementation of AI, which is a heavily subjective topic.

3.1. Qualitative Research

This qualitative research study aimed to explore how employees of small and medium-sized enterprises (SMEs) perceive the implementation of artificial intelligence (AI) in creative fields. The methodology involved semi-structured interviews with four participants, hereafter referred to as Interviewee 1, Interviewee 2, Interviewee 3, and Interviewee 4, to ensure anonymity and confidentiality. The selection of participants was conducted through purposive sampling, targeting employees engaged in creative roles within SMEs that had begun incorporating AI technologies into their workflows. This approach facilitated an in-depth understanding of their perspectives, concerns, and experiences related to AI integration.

Prior to the interviews, each participant received a research information letter that outlined the study's objectives, significance, and the nature of their involvement. This letter was designed to provide the interviewees with a comprehensive overview of the research topic, thereby enabling them to reflect on their experiences and formulate thoughtful responses during the interviews. By informing them in advance, the research aimed to foster a comfortable environment that would encourage candid discussions.

The interviews were conducted using online platforms, specifically Zoom and Microsoft Teams, utilizing voice calls to accommodate the participants' preferences and schedules. Each session lasted approximately 30 to 45 minutes, allowing ample time for in-depth exploration of the participants' thoughts and experiences regarding AI in their respective creative roles. With participants' consent, the calls were recorded to ensure accurate data collection and subsequently transcribed verbatim for analysis.

After completing the interviews, a consent letter form was sent to each participant. This document required their agreement to the use of their anonymized responses in the research. Participants were assured that their data would be handled with the utmost confidentiality and that their identities would not be disclosed in any published results. Following the completion of the transcription, the audio recordings were permanently deleted to further safeguard the participants' information, adhering to ethical research standards regarding data privacy and protection.

The transcribed interviews were compiled into a single document, which served as the primary data source for analysis. The analysis employed a thematic approach, whereby the transcripts were systematically reviewed to identify common themes and patterns related to the participants' perceptions of AI implementation in their work environments.

In addition to thematic analysis, the research also considered contextual factors influencing the participants' perceptions. This included the specific nature of their creative roles, the types of AI technologies being employed, and the organizational culture of their respective SMEs. By situating the findings within these broader contexts, the research aimed to provide nuanced insights into how employees in creative fields interpret and respond to the growing presence of AI in their workplaces.

Overall, this qualitative methodology facilitated a rich exploration of the employees' perceptions, capturing the complexities and nuances of their experiences with AI implementation in creative settings. The findings are intended to contribute to a deeper understanding of the implications of AI for job performance and satisfaction among SME employees, particularly in industries where creativity is paramount. The insights gleaned from this research will be instrumental in informing future discussions about the role of AI in the workplace and its impact on the human element of creative work.

3.2. Quantitative Research

This study utilized a quantitative research design to analyze the perceptions of employees in creative and marketing roles within small and medium-sized enterprises (SMEs) regarding the implementation of artificial intelligence (AI). The research process involved several key steps:

3.2.1. Dataset Selection and Preparation

The original dataset comprised 178 responses collected from employees across various industries and roles. These responses were gathered through a structured online survey designed to measure attitudes towards AI, including concerns, comfort levels, perceived job risks, and job satisfaction. To focus specifically on the creative and marketing sectors, the dataset was filtered to include only those respondents working in roles directly related to creative tasks, such as graphic design, content creation, advertising, and marketing management. This filtering process resulted in a final dataset of 28 responses, which provided a targeted sample for analysis.

3.2.2. Survey Instrument

The survey employed a mix of closed-ended and Likert-scale questions to quantitatively assess employee perceptions. Key constructs measured in the survey included:

- AI Concerns in Role: Assessed through statements that gauge respondents' worries about AI impacting their job roles and responsibilities (e.g., "I am concerned that AI will make my job obsolete").
- Comfort with AI in Work: Evaluated using questions that measure how comfortable respondents feel integrating AI tools into their daily tasks (e.g., "I feel comfortable using AI tools in my work").
- AI Job Risk: Explored through items focused on perceived job security and the potential risks associated with AI (e.g., "I believe that AI poses a risk to my job security").
- AI Impact on Job Satisfaction: Measured by evaluating how respondents believe AI implementation affects their overall job satisfaction (e.g., "AI positively impacts my job satisfaction").

3.2.3. Data Cleaning and Preparation

Prior to analysis, the dataset underwent a rigorous cleaning process to ensure data quality. This involved checking for and addressing any missing, inconsistent, or outlier responses. All categorical responses were encoded into numerical values for statistical analysis. For instance, Likert-scale responses (ranging from "Strongly Disagree" to "Strongly Agree") were transformed into numerical scores (1 to 5), allowing for the application of various statistical techniques.

3.2.4. Statistical Analysis

The analysis focused on examining relationships between key variables using T-tests and Analysis of Variance (ANOVA).

T-Tests: Paired T-tests were conducted to assess the differences between means of two related groups for selected pairs of variables. This allowed for the evaluation of direct relationships, such as:

- The relationship between AI concerns in one's role and AI's impact on job satisfaction.
- The relationship between AI concerns in one's role and comfort with AI in work.

ANOVA: One-way ANOVA was employed to determine if there were statistically significant differences among means when comparing more than two groups. For instance:

- Differences in job satisfaction levels based on varying degrees of AI concerns.
- The influence of AI concerns on comfort levels and job risk perceptions.

3.2.5. Significance Testing

A significance level of 0.05 was established for all statistical tests. P-values were calculated for each comparison to determine whether to reject or fail to reject the null hypothesis. A p-value less than 0.05 indicated a statistically significant relationship, while a p-value above 0.05 suggested no significant effect. Additionally, where appropriate, effect sizes were calculated to provide insight into the strength of the relationships observed.

4. Discussion

4.1. Qualitative research

Thematic Analysis of Employee Perceptions on AI in Creative Fields This thematic analysis examines employee perceptions of artificial intelligence (AI) in the creative field, based on interviews with four employees of small and medium-sized enterprises (SMEs), anonymized as Interviewees 1, 2, 3, and 4. The responses to the questions have been organized thematically, addressing each dimension of the study's hypotheses concerning the effects of AI on job performance, satisfaction, and creativity.

General Perception of AI

All four interviewees report varying degrees of AI usage, particularly for tasks that expedite mundane or repetitive aspects of their work. Interviewee 1 finds AI beneficial for simplifying routine tasks, such as separating objects from backgrounds in design software, which speeds up the workflow. Similarly, Interviewees 2 and 3 use AI for tasks like image editing and texture applications, noting that these tools reduce time but lack "life" or human touch in the results. Interviewee 4 describes AI as proficient in technical functions, like resizing or color adjustments, without enhancing the creative aspects of the work.

Across the interviews, there is a consensus that AI functions as a time-saving tool rather than a source of creative inspiration. The tool's utility lies in executing the technical or non-creative elements of projects, with Interviewees 2 and 3 emphasizing its supportive role in allowing them to allocate more time to creative components. The perception of AI as a facilitator rather than a creator establishes a baseline that highlights the employees' general acceptance of AI's efficiency in non-creative tasks.

AI and Job Performance

A primary theme concerning AI and job performance is ambivalence toward AI's impact. While AI makes tasks more efficient, it also brings a sense of disconnect between the employee and their work. Interviewee 1 reflects on AI's intrusion when searching for creative references, finding it an inconvenience due to the prevalence of low-quality AI-generated images that "lack soul." Interviewees 2 and 3 also express a sense of compromise, noting that while AI expedites tasks, the finished product feels incomplete or impersonal. Interviewee 4 suggests that, although AI accelerates the workflow, it diminishes the personal touch typically infused into their work.

The responses illustrate a paradox in AI's influence on job performance. Although AI optimizes technical efficiency, it detracts from the "hands-on" experience that the interviewees value, leading to dissatisfaction with the output. This theme indicates that AI's integration in creative roles may boost productivity but also risks undermining the authenticity and personal involvement that these employees associate with professional fulfillment.

Managerial Push for AI Integration

Interviewees reported varied experiences regarding managerial attitudes toward AI integration. Interviewee 1's manager, who has a creative background, encourages personal expression over AI use, fostering an environment that respects traditional artistic principles. Conversely, Interviewees 2 and 3 perceive a managerial push for AI due to its perceived efficiency benefits. This managerial expectation, especially among leaders without design backgrounds, pressures them to rely on AI, even when it may compromise quality.

Interviewee 4 represents a middle ground, reporting that while managers are curious about AI, they remain cautious and prioritize authenticity. This distinction suggests that managerial backgrounds and understanding of creative processes significantly influence whether AI is seen as an enhancement or threat to the artistic workflow. The responses indicate a potential misalignment between managerial expectations and employee perceptions of quality, revealing a friction point where AI efficiency might supersede creative quality.

AI and Creativity

The interviewees universally view AI as a non-enhancing force regarding creativity. Interviewee 1 specifies that AI neither hinders nor improves creativity; instead, it facilitates the efficiency of technical tasks, leaving the creative process to the individual. Interviewee 2 shares a similar sentiment, describing AI as merely a tool that assists in executing ideas rather than generating them. Interviewees 3 and 4 suggest that reliance on AI for technical elements might actually restrict their creativity, as they feel it removes opportunities to problem-solve and innovate on their own.

The interviews reveal a general sentiment that AI in creative fields serves more as a functional tool than a source of creative empowerment. This theme underscores a perceived boundary where AI can assist with technicalities but fails to contribute substantively to artistic ideation, reinforcing the human-centric nature of creativity in these roles.

AI and Job Satisfaction

The impact of AI on job satisfaction is notably mixed. Interviewees appreciate the convenience AI brings but also express concern over how AI minimizes the value of their skills. Interviewee 1, for instance, feels pride in their skills undermined when AI completes tasks in seconds that they trained for years to master. Interviewee 2 echoes this sentiment, noting that while clients are satisfied with AI-driven results, personal satisfaction is diminished due to the lack of personal input.

Interviewees 3 and 4 articulate a sense of frustration, as clients and managers may not discern the nuanced differences between AI-generated and manually created work. These findings suggest that while AI alleviates workload, it simultaneously diminishes professional pride and job satisfaction. AI's impact on job satisfaction, therefore, emerges as a double-edged sword, where employees must balance efficiency with a potential erosion of the skill-based value they place on their work.

Motivation and Professional Development

Most interviewees report no significant change in motivation, though Interviewee 4 notes a slight decrease, describing the work as "monotonous" due to repetitive interaction with AI. Interviewee 1, however, emphasizes the distinction between design and art, stressing that while AI may automate design tasks, it cannot replace the soulful elements inherent in true artistry.

Interviewees express a sense of skill devaluation as they find themselves correcting AI's mistakes rather than refining their own techniques. For example, Interviewee 3 worries about the potential atrophy of skills as AI automates more tasks. Interviewee 4, too, feels that AI complicates rather than enhances skill development, as the time saved by AI is offset by the frustration of rectifying its errors.

This theme highlights the perception that while AI aids efficiency, it may inadvertently impede skill mastery and diminish motivation, particularly among employees who view their profession as an ongoing craft.

Ethical Concerns and the Future of AI

All four interviewees express significant ethical concerns regarding AI's role in creative fields, particularly with respect to copyright and authenticity. Interviewee 1 condemns the replacement of human creativity with AI, emphasizing how generative AI models are often trained on copyrighted works without permission. Interviewees 2 and 3 further criticize the lack of regulation, noting how AI-generated works may deceive clients or the public about authenticity.

Looking forward, Interviewees envision an evolving yet cautiously optimistic role for AI. Interviewee 1 envisions increased reliance on AI tools by future designers who might lack foundational skills, leading to a potential dilution in industry standards. Interviewee 2 anticipates that AI advancements will maintain a noticeable gap from genuine artistry, while Interviewees 3 and 4 hope for educational programs that balance AI proficiency with traditional skills, underscoring the irreplaceable value of foundational techniques.

The ethical and future-oriented reflections in these interviews suggest an appreciation for AI's potential utility if applied within boundaries that preserve artistic integrity and transparency. The interviewees favor a balanced role for AI, supporting their creative work without compromising their unique artistic contributions.

Conclusion

This thematic analysis reveals a complex interplay between AI's time-saving advantages and its perceived undermining of the personal artistry valued by creative professionals in SMEs. While AI enhances efficiency, it does so at a cost to job satisfaction and creative authenticity, challenging employees to reconcile the convenience of automation with the loss of personal investment in their work. The mixed responses regarding managerial push indicate that acceptance of AI depends on management's understanding of the creative process, while ethical concerns underscore the urgency for regulatory frameworks to ensure AI complements rather than replaces human artistry.

In summary, the findings lean toward partial support for the alternative hypothesis (H1): employees of SMEs perceive AI's implementation in creative fields as significantly impacting job performance and satisfaction. However, the extent and nature of this impact are nuanced. Employees value AI's role in reducing workload but express reservations about its limitations, both ethically and creatively, indicating a preference for a controlled integration of AI that preserves the human elements at the core of creative work.

Figure 1. Word Cloud Highlighting Key Concepts Related to AI Implementation in SMEs that was done using code analyzing the interviews transcription.

4.2. Quantitative research

4.2.1. Hypotheses Testing

This study aimed to explore the perceptions of employees in small and medium-sized enterprises (SMEs) regarding the implementation of Artificial Intelligence (AI) in creative fields. We formulated two hypotheses:

- **Null Hypothesis (H0)**: Employees of SMEs perceive the implementation of AI in creative fields as having no significant effect on their job performance or job satisfaction.
- **Alternative Hypothesis (H1)**: Employees of SMEs perceive the implementation of AI in creative fields as having a significant effect (positive or negative) on their job performance or job satisfaction.

The analyses conducted revealed several significant relationships between employee perceptions of AI and various factors influencing job satisfaction and performance.

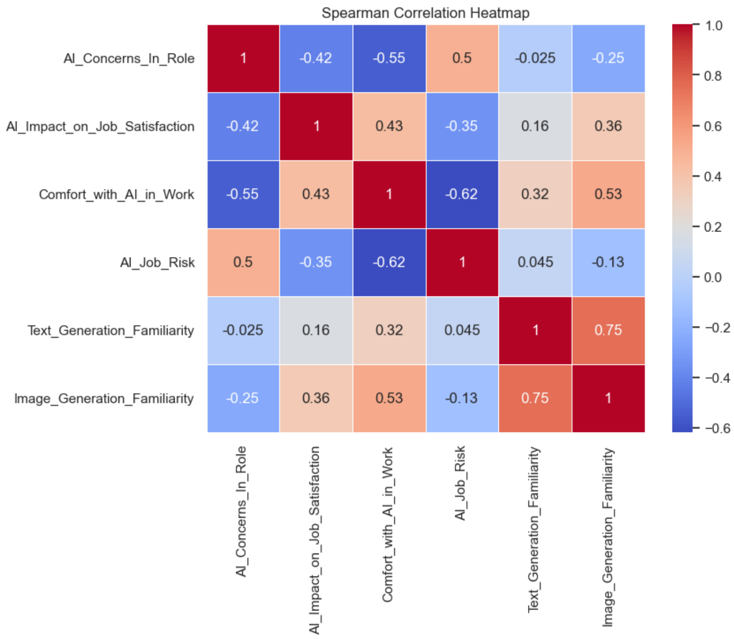

Figure 2. Spearman Correlation Heatmap showing relationships between perceptions of AI in SME roles.

The Spearman correlation heatmap reveals relationships between SME employees' perceptions of AI. There's a strong negative correlation (-0.55) between **AI Concerns in Role** and **Comfort with AI in Work**, showing that employees with AI concerns feel less comfortable with it. **AI Job Risk** positively correlates with **AI Concerns in Role** (0.5), suggesting higher job security concerns align with increased AI apprehensions. Familiarity with Image Generation tools correlates positively with **Comfort with AI in Work** (0.53), while **Text** and **Image Generation Familiarity** show a high correlation (0.75), highlighting familiarity patterns that may affect comfort and satisfaction with AI.

4.2.2. T-Tests Results

Comparison	T-statistic	P-value	Result	Interpretation
AI Concerns in Role vs. AI Impact on Job Satisfaction	-2.190473	0.037652	Reject H_0	Strong negative relationship
AI Concerns in Role vs. Comfort with AI in Work	-2.303502	0.029501	Support H_1	Negative relationship
AI Concerns in Role vs. AI Job Risk	2.388999	0.024440	Support H_1	Negative relationship
AI Impact on Job Satisfaction vs. Comfort with AI in Work	1.153501	0.259199	Fail to reject H_0	No significant relationship
AI Impact on Job Satisfaction vs. AI Job Risk	-1.534831	0.136906	Fail to reject H_0	No significant relationship
Comfort with AI in Work vs. AI Job Risk	-2.904666	0.007409	Reject H_0	Strong negative relationship

Table 1. Statistical Results for AI Concerns and Impacts on Job Variables

In summary, significant findings support H1, indicating that concerns about AI adversely affect job satisfaction and comfort while increasing perceived job risk. Furthermore, comfort with AI correlates with reduced job risk perceptions.

4.2.3. ANOVA Results Analysis

The ANOVA results further elucidate the relationships between the studied variables:

Comparison	F-value	P-value	Interpretation
AI Concerns In Role and AI Impact on Job Satisfaction	2.6299	0.0919	Variability in job satisfaction based on AI concerns; not significant at 5% level
AI Concerns In Role and Comfort with AI in Work	3.8485	0.0349	Significant impact of AI concerns on comfort levels; supports H1
AI Concerns In Role and AI Job Risk	4.1667	0.0274	Significant effect of AI concerns on perceived job risk; reinforces H1
AI Impact on Job Satisfaction and Comfort with AI in Work	2.9483	0.0419	Significant relationship; supports H1
AI Impact on Job Satisfaction and AI Job Risk	2.5677	0.0652	Potential effect; warrants further exploration
Comfort with AI in Work and AI Job Risk	5.5718	0.0048	Strong significant relationship; supports H1

Table 2. Summary of F and P values for AI Concerns and Job Factors

Overall, the findings suggest nuanced effects of employee perceptions regarding AI on job satisfaction, comfort levels, and perceived job risks. While concerns about AI do not significantly affect job satisfaction, they do impact comfort levels and risk perceptions. These results underscore the importance of addressing employee attitudes towards AI in creative fields and point to significant relationships that merit further investigation.

5. Conclusion

This research aimed to explore the perceptions of employees in small and medium-sized enterprises (SMEs) regarding the implementation of artificial intelligence (AI) in creative fields. Two hypotheses were formulated for this study: the null hypothesis (H0) posited that AI implementation has no significant effect on job performance or satisfaction, while the alternative hypothesis (H1) suggested that it does have a significant impact. Through a mixed-methods approach, combining both qualitative and quantitative analyses, the study gathered rich insights into employee attitudes and experiences regarding AI.

The qualitative findings highlighted the complex and often ambivalent relationship employees have with AI. While many acknowledged the efficiency gains AI offers, concerns about creativity, job satisfaction, and ethical implications emerged as predominant themes. Interviewees unanimously agreed that AI, while useful for repetitive tasks, does not enhance their creativity and can detract from their engagement and satisfaction with their work. The significance of human artistry and emotional depth in creative processes was emphasized, underscoring a perceived need for a balance between AI tools and human input.

Quantitative analysis supported the alternative hypothesis (H1), revealing significant relationships between employee concerns about AI and their job satisfaction and comfort levels. The results indicated that higher concerns about AI correlate with decreased comfort in using these technologies and increased perceptions of job risk. Although the influence of AI on job satisfaction was less clear-cut, the negative associations identified signal that employees' apprehensions about AI implementation warrant attention from management.

In summary, this study illustrates that while AI integration in creative roles can offer operational advantages, it also presents challenges that could undermine job satisfaction and creative engagement. As organizations continue to navigate the evolving landscape of AI technologies, fostering a supportive environment that values human creativity and addresses employee concerns will be critical in ensuring a harmonious coexistence of AI and human input in the creative process. Future research should further explore these dynamics to provide actionable insights for SMEs looking to implement AI thoughtfully and ethically [6].

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